Corporate Finance

Topic 1

Bernardo Ricca

References

- Main references:
  - Berk, J. and DeMarzo, P. (2011). Corporate finance, global ed. *Essex: Person Education Limited*
  - Tirole, J. (2010). *The theory of corporate finance*. Princeton University Press
- Complementary references:
  - Higgins, R. (2011). Analysis for financial management, 10e. *Burr Ridge: McGraw-Hill*
  - Brealey, R. A., Myers, S. C., Allen, F., and Mohanty, P. (2018). *Principles of Corporate Finance*. McGraw-Hill Education

Evaluation

- Final exam (**60%**)
- Midterm exam (**20%**)
- Groups of 2 students:
  - Problem sets (**10%**). The TA will randomly select one question to grade per problem set
  - Presentation of a paper (**10%**)

Financial economics

- **Asset pricing**: the goal is to understand the price of claims to uncertain future payments (e.g. stocks, derivatives, bonds)
  - Many assets (e.g. stocks and corporate bonds) are issued by corporations
  - Asset pricing models yield the appropriate risk-premium of a security $\Rightarrow$ essential for the calculation of appropriate discount rates in the valuation process

Financial economics

- **Corporate finance**: the goal is to understand some aspects of corporations (e.g. governance, board structure, CEO compensation) and how firms choose to finance their operations (e.g. stocks, bonds, bank debt, convertible bonds)
  - Corporate finance also studies how different financing choices affect the decisions of a firm and ultimately its value
- **Banking**: the goal is to understand the characteristics and impacts of banks and financial intermediaries more generally
  - Corporations contract loans and buy services from banks (e.g. issuance of financial contracts, financial and M&A advice)

Accounting review

- Goal: introduction to basic financial statements
- Key to measuring and understanding relevant aspects of firms
- Three main (financial) accounting statements:
  - **Balance sheet**: information about a firm's financial condition at a point in time
  - **Income (or earnings) statement**: information about a firm's financial performance over a period of time
  - **Cash flow statement**: information about the sources and uses of a firm's cash over a period of time

The balance sheet

- A balance sheet is a snapshot of the firm's financial position taken at a given point in time
- The firm's financial position includes all the assets the firm owns and all the claims against those assets
- It is divided into two sides: **assets (A)** on the left side and **liabilities (L)** and **shareholders' equity (E)** on the right
- Assets (A) summarize how the firm **uses its capital**, and the right side ($L + E$) summarizes the **sources of capital**

The balance sheet view of the corporation

Assets (A)	**Liabilities** (L)
"Resources that generate current and future cash flows for the corporation"	"Promises made to third parties"
	Equity (E)
	"Resources not promised away to third parties"

- Accounting identity: $A = L + E$
- The identity holds in **book value** or in **market value**
- In corporate finance we are mostly interested in **market values**

The balance sheet

- Every transaction preserves the accounting identity:
  - Purchase of a $1B plant on credit: assets: +$1B; liabilities: +$1B
- Which assets and liabilities are recognized?
  - In general, the more **tangible** the asset is, the easier it is to assign a value to it and thus the more likely it is to be recognized (e.g. cash, buildings, receivables)
  - **Intangible** assets such as patents, customer data, algorithms, and brand reputation are more difficult to measure and their recognition and valuation are treated more conservatively...
  - ... but intangible assets are increasingly important (some estimates: as much as 90% of S&P 500's total enterprise value)

The balance sheet

- How are assets and liabilities valued?
- **Acquisition (historical cost)**: the original price at which the asset was purchased (or the amount received for an obligation)
  - Different than the **market value**: the current market price for the asset or liability
  - Recent example: Silicon Valley Bank (held-to-maturity versus available-for-sale securities)
- Rules and regulations whether assets and liabilities are measured at acquisition of market value are complex and evolving...

The balance sheet view of the corporation

- Corporations sell two things
- **(i) Goods and services**
  - Firms invest in assets (property, plant, and equipment or PP&E, working capital, etc.) and hire workers that enable those sales
  - The sales generate current and future cash flows that underpin the market value of its assets
- **(ii) Promises**
  - Firms must raise money to invest in their assets
  - They are able to raise money by promising to outside investors part of the future cash flows generated from their sales

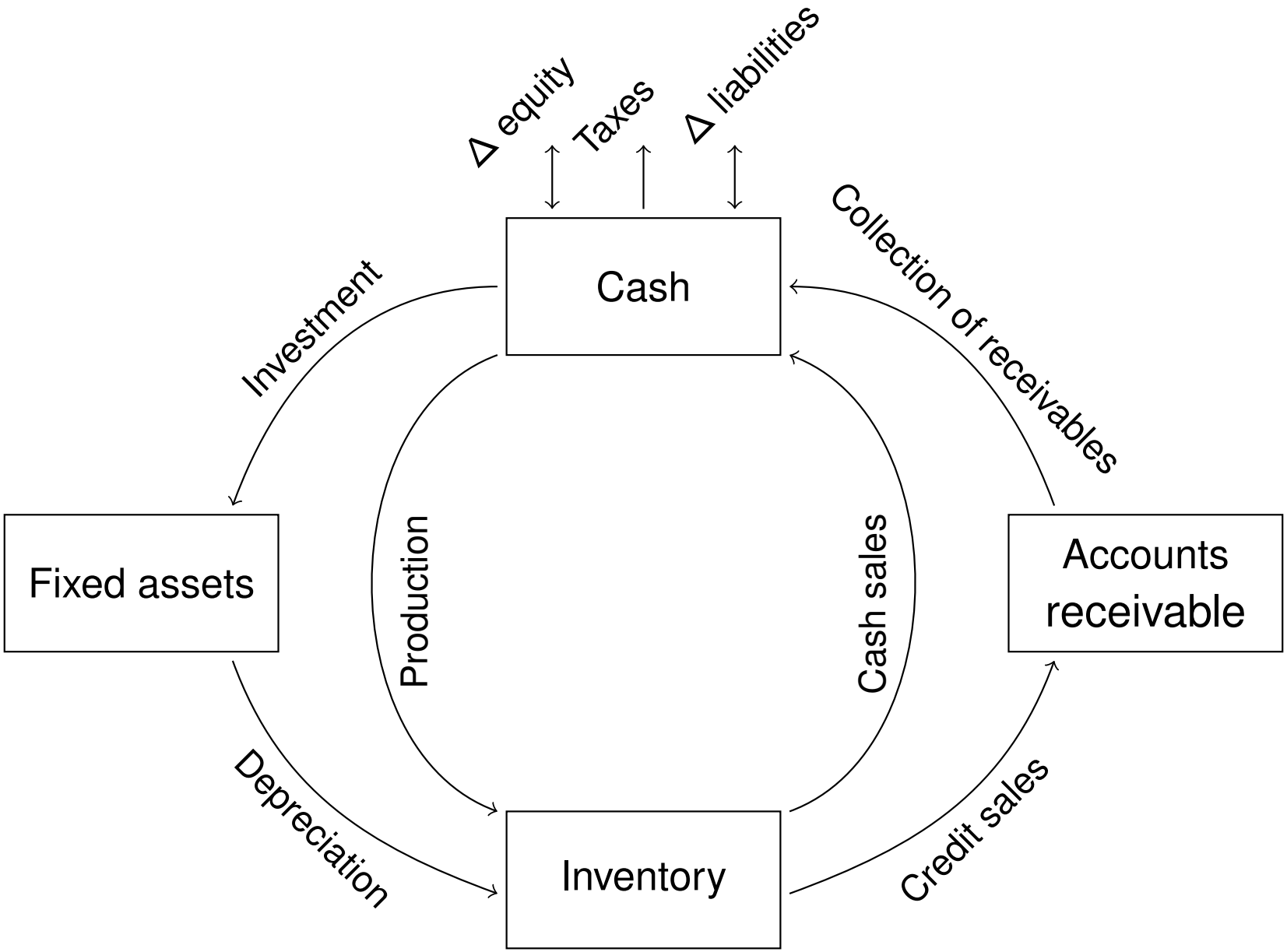
Promises

- There are two types of promises
- **Liabilities** or **promises to non-owners (creditors)**: senior (e.g. bonds, bank loans, payables)
- **Equity** or **promises to owners (shareholders)**: junior (e.g. stocks)
- Owners are residual claimants: they receive future cash flows that are not promised away to non-owners (creditors)
- The choice between different types of promises is a key question in corporate finance

The balance sheet view of the corporation

- The fact that cash flows generated from assets are split into payments to owners and non-owners explains the accounting identity in market value
- Recall that the value of an asset is the expected discounted value of its future payments
- $FCF_A = FCF_L + FCF_E \quad \Longrightarrow \quad A = L + E$

The cash flow-production cycle



Source: Higgins (2011)

Financial statements: a quick review

- We will closely follow Berk and DeMarzo (2011)
- **Current assets**: assets that can be converted into cash within one year (liquid assets)
- Examples:
  - Cash and other marketable securities: short-term, liquid, and low-risk investments (money market investments, government debt, bank deposits)
  - Accounts receivable: amount owed by customers who have purchased goods and services on credit
  - Inventories: raw materials, work-in-progress, and finished goods

Assets

- **Non-current assets**: “long-term” assets. Examples:
  - Fixed assets such as property, plant, and equipment
  - Goodwill: the difference between the market and the fair value (the accountant’s estimate of the potential value of an asset, not necessarily equal to book value) of an acquisition
- Over time, fixed assets are worn out. The accountant reduces the book value of fixed assets by an amount known as **depreciation**
- Depreciation is not a cash expense
- The book value of an asset: acquisition cost minus accumulated depreciation

Assets: rise in corporate cash

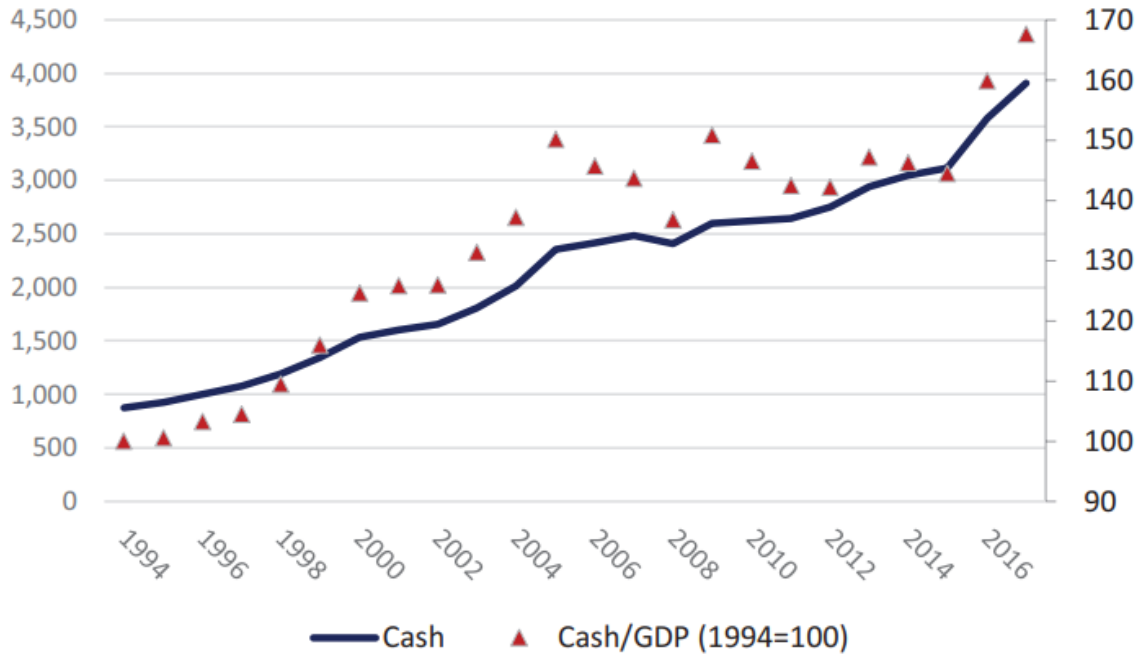


Figure 1
Aggregate corporate cash: Flow of funds
The data is from the Federal Reserve Flow of Funds, Table L. 102 Nonfinancial Business (June 1, 2018). We summed rows 2 through 11. This includes checking and savings accounts of non-financial businesses as well as investment in debt securities (e.g., commercial paper, government bonds, and loans). The data is from 1994 to 2017.

Source: Faulkender et al. (2019)

Liabilities

- **Current liabilities**: obligations that are due within one year
- Examples:
  - All debt payments due in the next year (from short-term as well as long-term debt instruments)
  - Accounts payable: amounts owed to suppliers for products or services purchased with credit
  - Accrual items such as salary and taxes that are owed but have not yet been paid
  - Deferred or unearned revenue: cash that has been received for products that have not yet been delivered

Liabilities

- **Non-current liabilities**: obligations that extend beyond one year (“long-term”)
- Example: debt payments that are not due within the next accounting year

Net working capital

- The difference between current assets and current liabilities is the firm's **net working capital (NWC)**, or simply working capital
- It is a measure of the firm's financial health in the short-term: firms with low net working capital might not have enough resources to meet short-term obligations

Example: GPA

Table: Balance sheet as of 31/12/2020 (in millions of BRL)

Assets		**Liabilities & shareholders' equity**	
Current assets	17,641	**Current liabilities**	18,483
Cash and equivalents	8,711	Trade payable, net	11,424
Trade receivable, net	686	Borrowings and financing	2,309
Inventories	6,536	Lease liabilities	947
Other	1,708	Financing of property and equipment	100
		Other	3,703
Non-current assets	35,654	**Non-current liabilities**	18,005
Property and equipment, net	19,888	Borrowings and financing	6,842
Intangible	6,164	Lease liabilities	7,427
Other	9,602	Other	3,736
		Total liabilities	36,488
		Shareholders' equity	16,807
Total assets	53,295	**Total liabilities & shareholders' equity**	53,295

- Remark: note the importance of suppliers in the provision of short-term financing (trade payable)

Shareholders' equity

- Difference between assets and liabilities: **book value of equity**
- Accounting measure of the net worth of the firm
- The book value of equity differs from the actual amount that investors are willing to pay for the equity portion of the firm, known as the **market value** or **market capitalization** or **market cap**
- Market value of a publicly traded company: share price times the number of common shares outstanding
- **Market values are forward-looking** (they reflect the expected discounted value of future payoffs to shareholders), while **book values are backward-looking**

Shareholders' equity

- Examples of drivers of the difference between book and market values
- The values of most of the assets on the balance sheet are computed using historical costs rather than their current market value
  - Historical costs do not incorporate changes to the future income the assets might generate
  - Only a few assets and liabilities are actively traded and can be marked to market
- Many of the firms' valuable assets (or liabilities) do not appear on the balance sheet:
  - Assets: quality of the employees and the management team, value of future R&D innovations, reputation, loyal customers; liabilities: pending lawsuits (e.g. regarding product liability claims)

Example: Starbucks

Table: Balance sheet as of 27/12/2020 (in millions of USD)

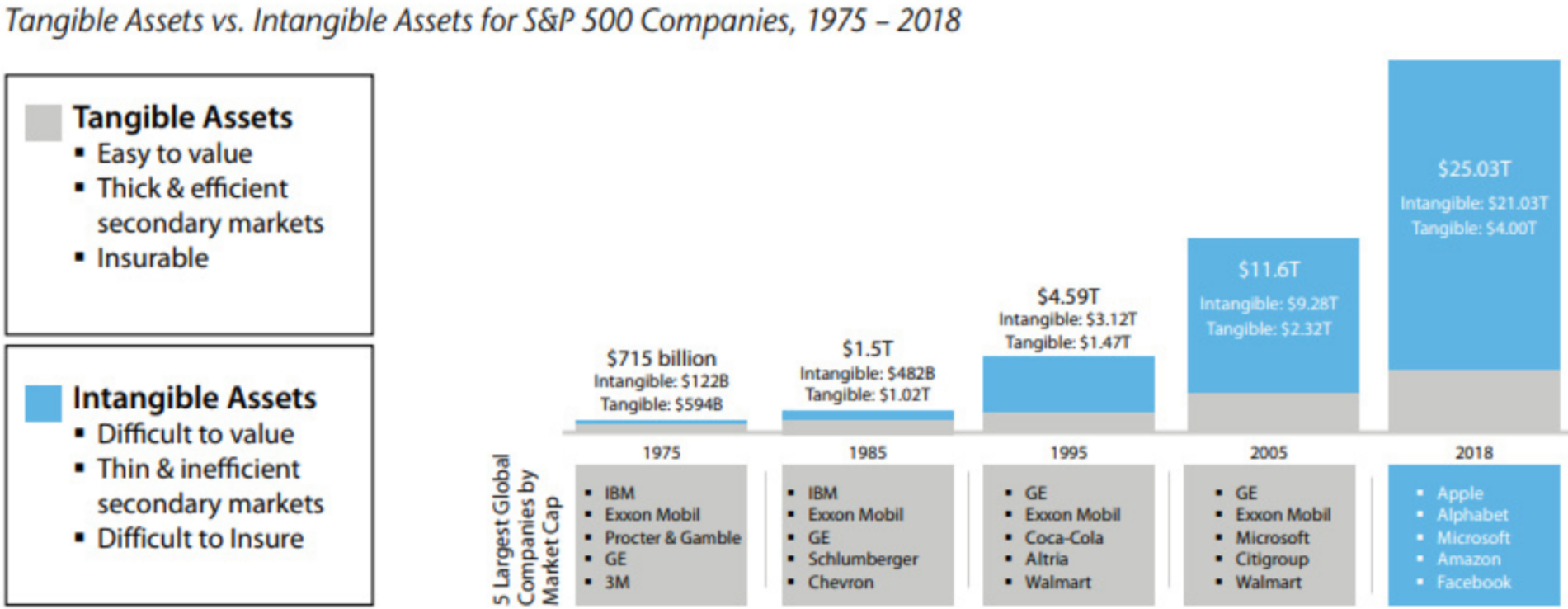
Assets		**Liabilities & shareholder's equity**	
Current assets	8,358	**Current liabilities**	7,884
Cash and equivalents	5,028	Accounts payable	1,051
Accounts receivable, net	888	Accrued payroll and benefits	685
Inventories	1,472	Current portion of operating lease liability	1,268
Other	970	Short-term debt	493
		Current portion of long-term debt	750
		Stored value card liability and current portion of deferred revenue	1,871
		Other	1,767
Non-current assets	21,611	**Non-current liabilities**	29,989
Property and equipment, net	6,118	Long-term debt	14,674
Operating lease, right-of-use asset	8,199	Operating lease liability	7,755
Goodwill	3,707	Other	7,561
Other	3,587	**Total liabilities**	37,872
		Shareholders' equity (deficit)	(7,904)
Total assets	29,968	**Total liabilities & equity**	29,968

- Negative book value of USD 7.9B

Example: Starbucks

- Is Starbucks insolvent?
- No. Negative book value of USD 7.9B, yet the market capitalization was 122.6B on Dec 28
- This example highlights the importance of using market values

Intangible assets



Source: https://www.visualcapitalist.com/intangible-assets-driver-company-value/

Market-to-book ratio

- **Market-to-book ratio**: market cap divided by the book value of equity
- Growth stocks: high market-to-book ratio
- Value stocks: low market-to-book ratio

Debt-equity ratio

- **Debt-equity ratio**: total liabilities divided by shareholders' equity
- We can compute this measure using either book or market values for equity and debt
- It is a measure of the firm's leverage
- The market debt-equity ratio is more informative about the firm's leverage and capacity to meet future debt payments
  - Even firms with negative book equity (recall the Starbucks example) might be able to contract a lot of debt because they are expected to be profitable in the future
  - Particularly useful for assessing the leverage of high-growth and start-up firms

Debt-equity ratio

- It is common to use the book value of debt as a proxy for the market value of debt
- It is also common to replace debt with **net debt**, defined by liabilities less cash and marketable securities
  - It is a better way to measure the leverage of companies with large excess balances of cash
  - We can view cash and marketable securities as negative debt: they can be used on short notice to repay debts

Enterprise value

- Proxy for the value of the business as a whole, or the value of the business assets of the firm
- Enterprise value = Market equity value + Debt - Cash
- How much cash does an investor need to be the single owner of all the assets of a firm?
- Total cash required to buy all equity and pay off the debts, but subtracting the cash that the firm has (the investor can use this cash to pay off the debt for instance)

Current ratio

- **Current ratio**: ratio of current assets to current liabilities
- **Liquidity ratio** (or quick ratio or acid test): ratio of current assets other than inventory to current liabilities
- It is a measure of the firm's ability to meet short-term obligations
- Some **debt covenants** require the current ratio to be above a certain value

Income statement

- The **income statement** or **profit and loss statement (P&L)** attempts to measure the profits the firm generates during a period of time
  - It lists the firm's revenues and expenses over a period of time
  - Revenues increase the book value of equity, while expenses reduce it
- Accountants apply the **accrual principle**: revenue is recognized at the time of the sale, not when the customer pays. Similarly, expenses are recorded when goods and services are purchased, not when suppliers are paid
- The accrual method matches revenues and expenses in the period in which the transaction occurs (the **matching principle**)

Income statement: accrual principle

	Cash flow?	Revenue / expense?
A customer makes a purchase, and pays in cash	Yes	Yes
A customer pays the bill for a purchase they made on a previous date	Yes	No
A customer makes a purchase, but agrees to make the payment later	No	Yes

Income statement

- The accountant does not charge the full cost of a long-term fixed asset in the year of the investment
  - The profit would be too low in the year of the investment, and too high in the other productive years of the asset (as the asset contributes to revenues but not to expenses)
- The solution is to spread the cost of the asset over its productive life in the form of **depreciation**
- It is a way to properly match revenues and expenses during a period
- Depreciation is not a cash expense; the cash outlay only occurs at the time of the investment

Income statement

- The amount of depreciation depends on the length of the productive life of the asset, its salvage value, and the method of allocation (straight line, accelerated)
- Also following the matching principle, accountants do not record **increases in inventories** as an expense

Income statement

- **Gross profit**: revenues from sales of products minus costs incurred to make and sell the products (**costs of goods sold** or **cost of sales**)

- **Other operating expenses**: expenses from the ordinary course of running the business that are not directly related to producing the goods being sold (that is, other than costs of goods sold). Examples: administrative expenses and overhead, salaries, marketing costs, R&D expenses, depreciation

- We can assume that most of the expenses in costs of goods sold are **variable** (change as sales vary), while most of the other operating expenses are **fixed** (independent of the amount of sales)

  - Companies with a high proportion of fixed costs are more vulnerable to sale declines (high **operating leverage**)

Income statement

- **Operating income**: revenues net of operating expenses
- **Earnings before interest and taxes (EBIT)**: operating income plus any net income related to activities that are not a core part of the firm's business (**nonoperating** income)
  - It is a measure of the firm's income before it is split among debtholders, equityholders, and the tax authorities
- **Net income** or (net) earnings or (net) profits or bottom line: EBIT minus interest expenses and taxes
  - Apart from any equity issuance or payout to equityholders, net income determines the growth in book equity

Statement of equity

- Shareholder equity = [paid in capital + retained earnings]
- The statement of equity or statement of changes in equity reports the changes in the firm's equity
- [Beginning book equity + **net income** - dividends +/- other changes] = Ending book equity
- Other changes include issuance of equity, treasury share transactions, stock-based compensation

Example GM

Table: Income statement as of 31/12/2020 (in millions of USD)

Net sales automotive	108,673
Net sales Financial	13,812
Net sales	122,485
Automotive and other cost of sales	97,539
GM financial interest, operating and other expenses	11,274
Cost of sales	108,813
Gross profit	13,672
Automotive and other selling, general, and administrative expenses	7,038
Operating income	6,634
Nonoperating income	2,559
EBIT	9,193
Interest expense	1,098
Pre-tax income	8,095
Taxes	1,774
Net income	6,321

Income statement analysis

- **Earnigs per share (EPS)**: net income divided by the total number of shares outstanding
- If the company compensates employees with stock options or if the firm has issued convertible bonds, the number of outstanding shares might increase
- This growth in the number of stocks is known as dilution
- **Diluted (EPS)**: net income divided by the total number of shares assuming that stock options have been exercised and convertible debt has been converted outstanding

Income statement analysis

- **Gross margin**: ratio of gross profit to sales
  - It is a measure of the ability to sell a product for more than the direct cost of producing it
  - It also measures the contribution of sales to the payment of fixed costs and profits
- **Operating margin**: ratio of operating income to sales. This measure includes indirect operating costs
  - It is a measure of the ability to sell a product for more than the direct and indirect cost of producing it
  - It is a measure of the efficiency of the firm's operations
  - But it is also affected by the ability of a firm to charge higher mark-ups/prices (market power)

Economic earnings

- Does a positive net income imply that equityholders are properly compensated?
- Equityholders make a profit if they obtain a return larger than the required rate of return on equity (hurdle rate)
- The required rate of return takes into account the time value of money and the amount of risk of the investment

Income statement analysis

- **Net profit margin**: ratio of net income to sales
  - Share of each dollar of sales that is available to shareholders after the firm pays its expenses (including interest expenses) and taxes
  - It is also a measure of the efficiency of the firm's operations
  - But it can be affected by the capital structure of the firm (e.g. a highly leveraged firm pays a lot of interest to creditors)

Income statement analysis

- **Accounts receivable days**: ratio of accounts receivable to average daily sales
  - It is a measure of the amount of time to collect payments from customers
- **Accounts payable days**: ratio of accounts payable to average daily purchases

Income statement analysis

- **EBITDA**: earnings before interest, taxes, depreciation and amortization
  - As depreciation and amortization (depreciation of intangible assets) are not cash expenses, it is a better measure of the cash the firm obtains from its operations
- **Interest coverage ratio**: ratio of EBIT to interest expenses
  - It is a measure of the firm's current ability to meet interest payments
  - Companies with the same leverage can have different coverage ratios due to differences in profitability
  - Companies with low coverage ratios might still be able to borrow because they are expected to generate large cash flows in the future (e.g. start-up, rapidly growing firms)

Income statement analysis

- **Return on equity (ROE)**: ratio of net income to the book value of equity
  - It is a measure of the return the firm has earned on its past equity investments
  - But it can also be affected by the capital structure of the firm
- **Return on assets (ROA)**: ratio of net income to the book value of assets
- **Asset turnover**: ratio of sales to the book value of assets
  - It is a measure of how efficiently the firm is using its assets to generate sales

The DuPont identity

- The DuPont identity decomposes ROE into

$$\text{ROE} = \underbrace{\underbrace{\left(\frac{\text{Net income}}{\text{Sales}}\right)}_{\text{Net profit margin}} \times \underbrace{\left(\frac{\text{Sales}}{\text{Total assets}}\right)}_{\text{Asset turnover}}}_{\text{Return on assets (ROA)}} \times \underbrace{\left(\frac{\text{Total assets}}{\text{Total equity}}\right)}_{\text{Financial leverage}}$$

- ROE increases if (other things constant)
  - Earnings obtained for each dollar of sales increase (net profit margin)
  - Sales generated for each dollar of assets increase (asset turnover)
  - The amount of equity used to finance the assets decreases (or financial leverage increases)
- Should managers maximize financial leverage?

The DuPont identity

Table: Determinants of ROE, 2013

	ROE	Net profit margin	Asset turnover	Financial leverage
Google	14.80%	21.60%	0.54	1.27
JPMorgan	8.50%	18.60%	0.04	11.44
Hyundai Motor	15.90%	10.30%	0.65	2.36
Chevron	14.40%	9.70%	0.87	1.7

Source: Higgins (2011)

- Net profit margin and asset turnover tend to vary inversely: companies that add a lot of value to a product and thus can charge higher prices usually require lots of assets

ROE as a measure of performance

- Is a company with a high ROE better than a company with a smaller one? Does an increase in ROE mean the firm's performance has improved?
- Not necessarily. ROE is backward-looking and focused on a single year
  - It would fall after an equity issuance to fund a project because the investment takes a while to turn into profits
- ROE ignores risk: ROE might be high simply because the business is very risky
- ROE measures the return over the book value of equity, not the market value (the current value of shares)

Return on invested capital (ROIC)

- Also known as return on net assets (RONA), ROIC is defined by

$$\text{ROIC} = \frac{\text{EBIT(1 - Tax rate)}}{\text{Interest-bearing debt + Equity}}$$

- This measures controls for some of the distortions of ROA and ROE. Example with two identical companies

	Company A	Company B
Debt (10% interest)	900	0
Equity	100	1000
EBIT	120	120
Interest expense	90	0
Earnings before tax	30	120
Tax (40%)	12	48
Net income	18	72
ROE	18%	7.2%
ROA	1.8%	7.2%
ROIC	7.2%	7.2%

Source: Higgins (2011)

Valuation ratios

- **Price-earnings ratio (P/E)**: ration of market capitalization to net income

$$\text{P/E ratio} = \frac{\text{Market capitalization}}{\text{Net income}} = \frac{\text{Share price}}{\text{Earnings per share}}$$

- Better growth prospects increase the P/E ratio, while increases in risk decrease it
- P/E ratios can vary over time and across industries
- It can be useful to make intra-industry comparisons of how firms are priced in the market: a high P/E ratio (relative to peers in the same industry) could indicate that the firm is overvalued
- P/E ratios are also influenced by the capital structure (leverage)

Statement of cash flows

- The income statement measures the firm profit over a period of time
- But it does not indicate the amount of cash the firm has generated:
  - The income statement has non-cash entries such as depreciation and amortization
  - Some cash expenses, such as capital expenditures or expenditures on inventory, are not reported on the income statement
- In other words, profits do not equal cash flow
  - It is possible that a profitable firm has not enough cash to meet its obligations

Statement of cash flows

- The statement of cash flows reports how much cash flow a firm has generated during a given period and how that cash flow has been allocated
- It measures how the firm's cash balance changes over time
- Investors are ultimately interested in the amount of cash flows they can extract from the business

Statement of cash flows: operating activities

- The statement of cash flows is divided into three sections
- **Operating activities**: it starts with net income and adds back all non-cash entries (e.g. depreciation, deferred taxes) related to the firm's operating activities
- **Depreciation**: it is deducted when computing net income but it is not a cash expense. Therefore, we add it back to net income to measure the amount of cash the firm has generated
- Same with other non-cash expenses such as **amortization** and **accrued items** such as salaries and taxes

Statement of cash flows: operating activities

- **Accounts receivable**: sales are recorded as part of net income even when the payment has not been received. We must adjust cash flows by deducting the increases in accounts receivable
- Similarly, we add back increases in **accounts payable**: purchases from suppliers that have not been paid yet
- We deduct increases to **inventory**, as they are not recorded as an expense in the income statement

Statement of cash flows: investing activity

- Purchases of property, plant, and equipment (capital expenditures) are not recorded in the income statement
- Instead, the the firm depreciates (consumes) these assets and deducts depreciation over time
- We have to subtract capital expenditures

Statement of cash flows: financing activity

- It includes cash outflows to equityholders such as dividend payments and share repurchases
- It also includes changes in borrowing, such as issuance of new debt and principal payments

Example: GM

Table: Statement of cash flows as of 31/12/2020 (in millions of USD)

Cash flows from operating activities	
Net income	6,321
Depreciation and amortization	12,815
Accounts receivable	(1,341)
Accounts payable	42
Inventories	(104)
Other adjustments	(1,063)
Net cash provided by operating activities	**16,670**
Cash flown from investing activities	
Expenditures for property	(5,300)
Available-for-sale marketable securities, acquisitions	(16,204)
Available-for-sale marketable securities, liquidation	11,941
Other adjustments	(12,263)
Net cash used in investing activities	**(21,826)**
Cash flow from financing activities	
Net increase (decrease) in short-term debt	277
Debt issuance (original maturities > 3 months)	78,527
Debt payments (original maturities > 3 months)	(72,663)
Stock issuance	492
Dividends paid	(669)
Other financing activituies	(412)
Net cash provided by (used in) financing activities	**5,552**
Foreign exchange adjustment	(222)
Net increase in cash and cash equivalents	**174**
Cash and cash equivalents at the beginning of period	22,943
Cash and cash equivalents at end of period	23,117

Measures of cash flow

- **Net cash flow** or cash earnings: net income + noncash items
- **Cash flow from operating activities**: net cash flow +- changes in current assets and liabilities
  - +- changes in current assets and liabilities or - changes in net working capital
  - On the cash flow statement, net cash provided by operating activities
  - The amount of cash generated by the firm's core operations over a period of time
- Δ cash= net income - Δ noncash assets + Δ liabilities + Δ paid in capital - dividends

Measures of cash flow

- **Free cash flow (FCF)** to the firm or **unlevered free cash flow**: total cash available for distribution to owners and creditors after funding all investment activities:

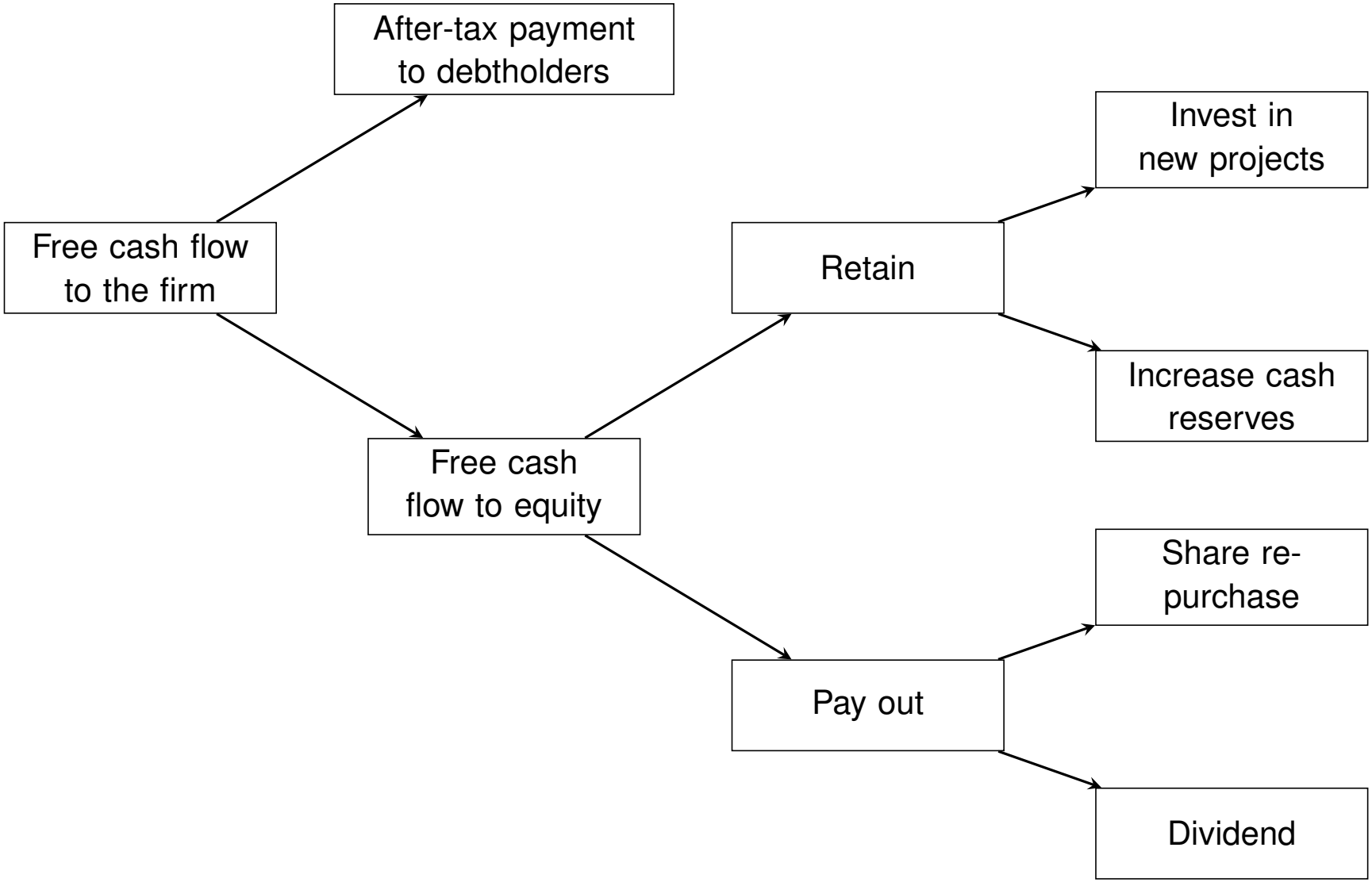
  FCF = EBIT(1 - Tax rate) + Depreciation - Capital expenditures - Working capital investments

  - EBIT(1 - Tax rate): income the company earns regardless of how the business is financed, that is, excluding interest expenses associated with debt financing (unlevered or all-equity)
  - Depreciation (and any other noncash items) are not cash outlays
  - Working capital investment: increase in current assets necessary to support the firm's operations, minus any increase in non-interest bearing current liabilities

Measures of cash flow

- **Free cash flow to equity (FCFE)** or **levered free cash flow**: total cash available for distribution to equityholders or for reinvestment in the business after meeting interest payments
  - Essentially, cash flow from operating activities minus capital expenditures
  - Or FCF minus (after tax) interest expenses
  - We should also include net debt issuance (or deduct net debt repayment) if the firm is changing its level of debt
- Forecasts of **FCF** are used to compute the market **enterprise value**, while forecasts of **FCFE** are used to determine the market **equity value**

Uses of free cash flow to the firm



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